

OMM Lab: Area of Greatest Restriction (AGR), Facilitated Positional Release (FPR) for the Spine

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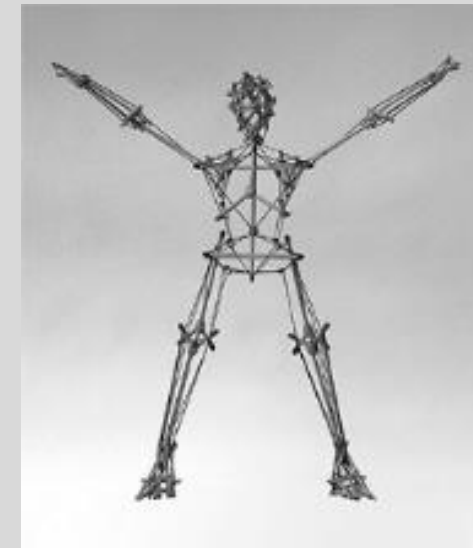
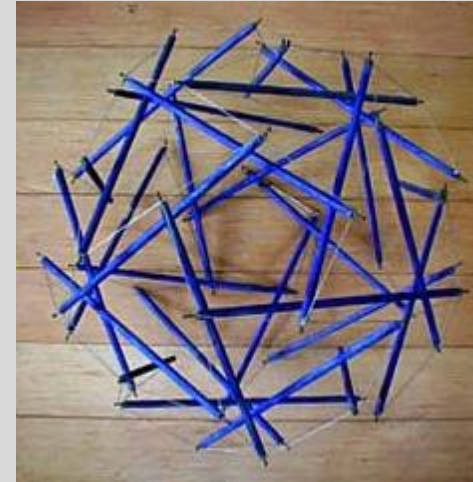
Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Lab Session Objectives

1. Students will understand the application and approach to evaluate a patient using the Area of Greatest Restriction (AGR).
2. Students will be able to perform screening finding the AGR for the upper half and lower half of the body.
3. Students will apply osteopathic treatment to the regions of AGR and understand how to sequence their treatment approach.
4. Student will review the history and principles of FPR.
5. Students will learn to diagnose and treat cervical, thoracic and lumbar SD with FPR.

Concept of Tensegrity in OMM

- Tensegrity = structural integrity via balance of **tension** and **compression**.
- **Body as a biotensegrity system:**
 - Rods = bones (compression).
 - Wires = muscles, fascia, ligaments (tension).
- Movement or dysfunction in one area changes forces throughout the body → informs whole-person assessment.



Mechanobiology and Diseases of Mechanotransduction

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- Mechanical forces regulate:
 - Cellular biochemistry & gene expression.
 - Tissue development & repair.
- Disruption in forces may lead to:
 - Altered posture, circulation, nerve function.
 - Disease at the tissue/cellular level.



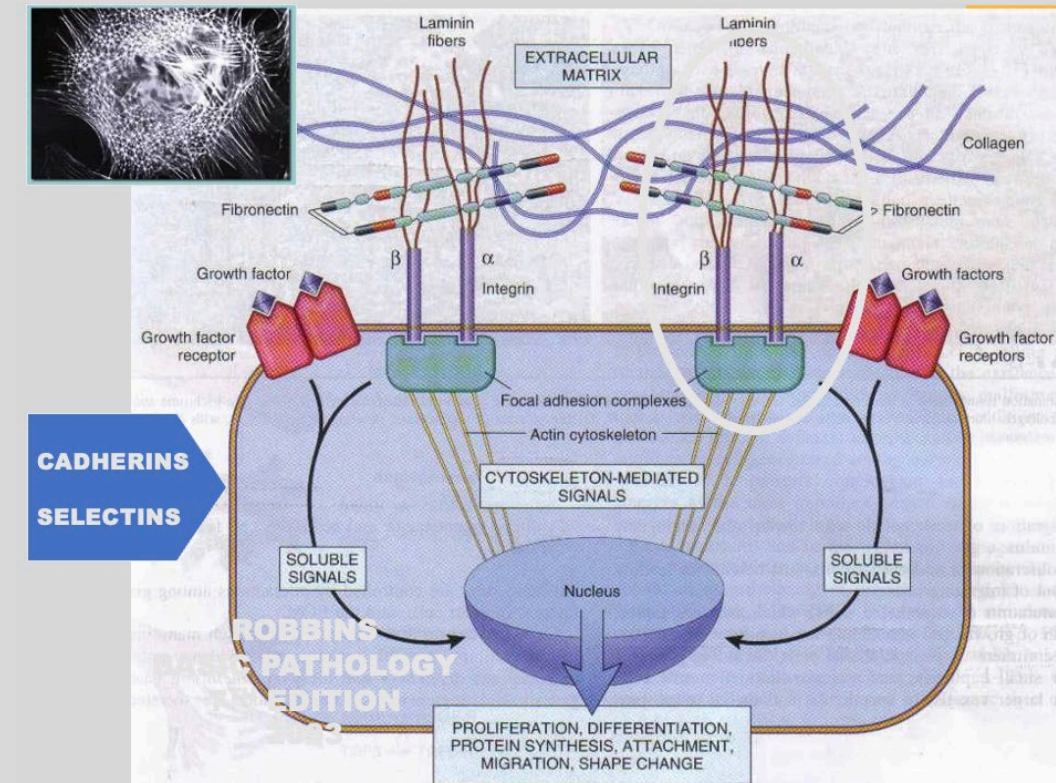
Mechanobiology and Diseases of Mechanotransduction

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- Mechanical forces are critical regulators of cellular biochemistry and gene expression as well as tissue development.
- **There is an “undeniable physical basis of disease.”**
- **Physical interventions can influence cell and tissue function.**
- Mechanical forces serve as important regulators at the cell and molecular levels, and they are equally potent as chemical cues.
- **“Forces that are applied to the entire organism (e.g., due to gravity or movement) would be distributed to individual cells via their adhesions to the ECM support scaffolds that link cells and tissues throughout the body.”**



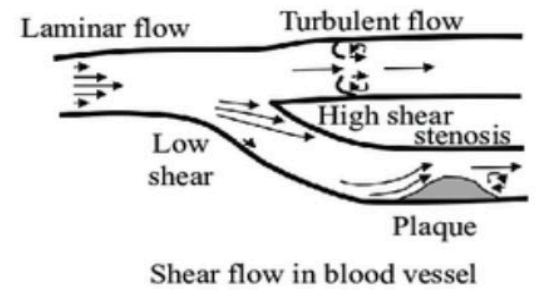
Mechanobiology and Diseases of Mechanotransduction

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- Mechanical forces directly regulate the shape and function of essentially all cell types.
- The global shape of the cell dictates its behavior (e.g., growth versus differentiation or apoptosis), and these effects are mediated through tension- dependent changes in cytoskeletal structure and mechanics.
- Manual type therapies have known the therapeutic value because they alter mechanotransduction.
- Most of the clinical problems that bring a patient to the doctor's office result from changes in tissue structure and mechanics.
- Abnormal cell and tissue responses to mechanical stress may actively contribute to the development of many diseases and ailments.
- **Consequently it may be wise to look for a physical cause for disease.**

Additional examples of mechanical effects on the human body

- Increased bone formation vs cartilage to mechanical loading
- Increased skeletal mass to stretch
- Increased hormone secretion of atrial natriuretic factor by heart cells to stretch
- Stretching of lung tissue regulates the synthesis of extracellular matrix
- Exposure to microgravity is associated with atrophy in heart, muscle, and bone, which is also frequented in aging
- Shear flow in blood vessel influences physiology and pathophysiology of endothelial cells and blood cells
- Increased growth of skin, bone, fibroblast to stretch.
- Kidney, bladder cells, pulmonary epithelial cells changing in growth and secretions to stretch.
- Increased deposition of bone to mechanical stress

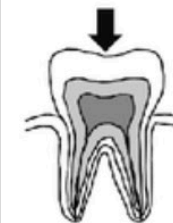


Contraction, compression, stretching

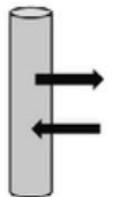


Stretching

Gravity



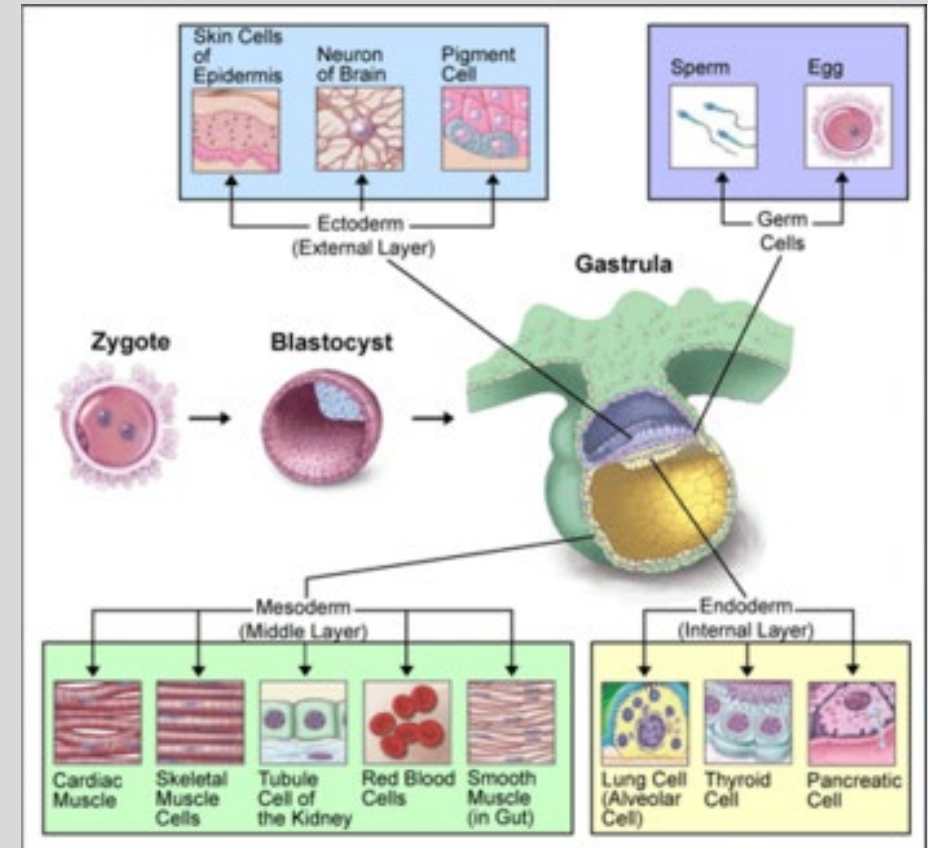
Compression



Hydrostatic, osmotic pressure

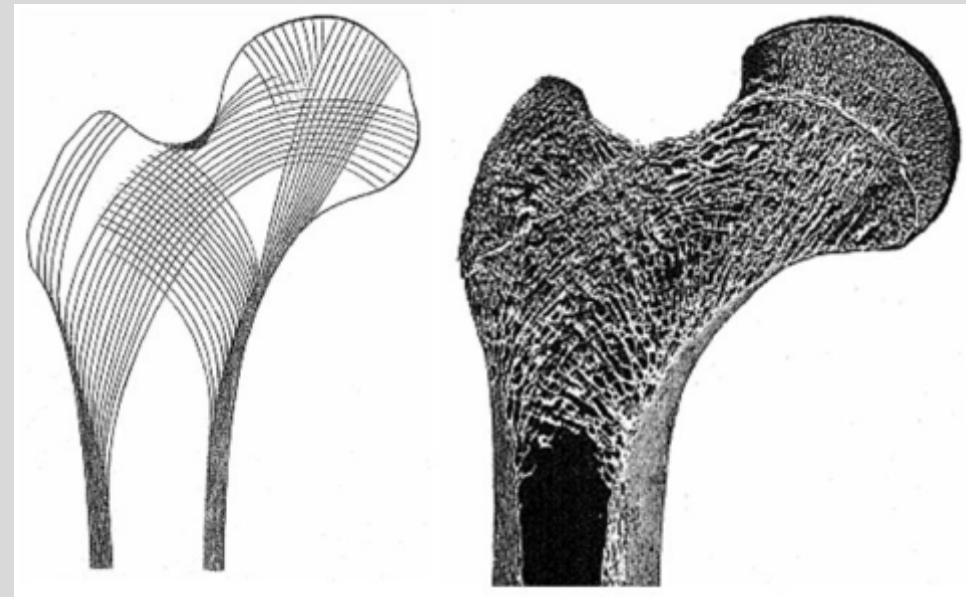
Embryology

- Biokinetics (movements) or Biodynamics (forces) of human development and stems mainly from work done at Göttingen University by the embryologist E.Blechschildt
- Biomechanical forces of the embryological heart trigger the production of chemicals which in turn trigger the cellular formation of hematopoietic cells which differentiate into blood.



Wolff's Law in Bone

- Dr. Julius Wolff in 1892 observed that bone is reshaped in response to the forces acting on it.
- Structure and shape of bone adapt to the loading conditions.





Piezoelectric Effect

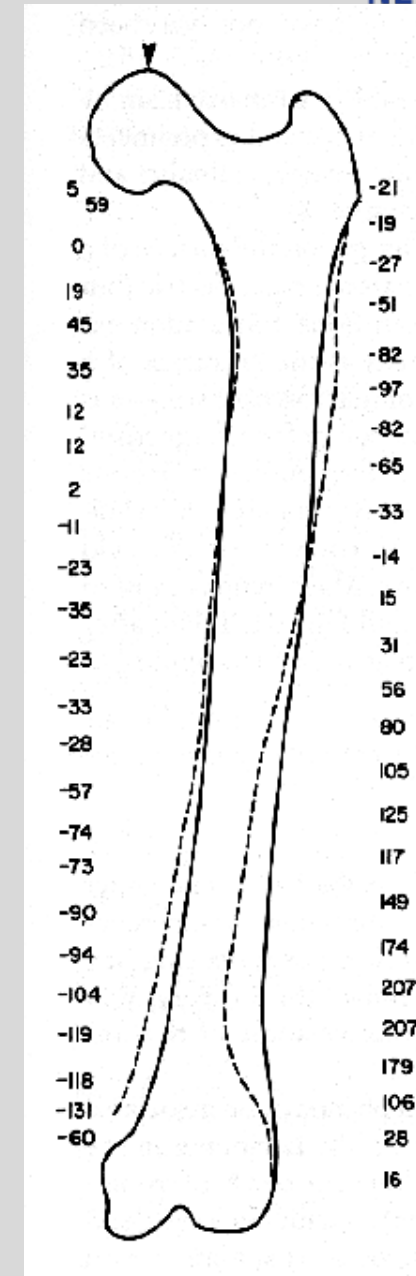
- Mechanical stress results in electric polarization.
- Piezoelectric properties of bone is a hypothesized feedback mechanism for bone remodeling.
- Increased deposition of bone to due to mechanical stress.

<http://www.biotele.com/EL/EL4/Piezo.html>

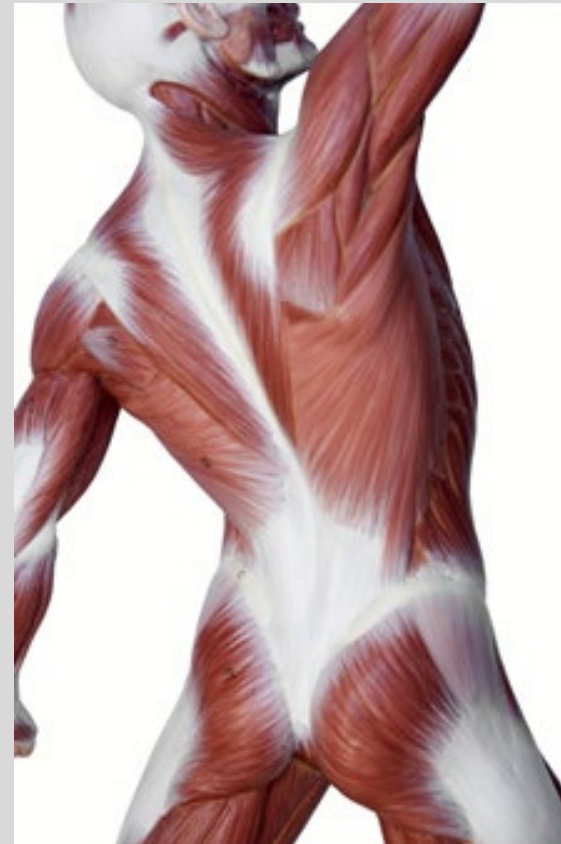
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Connection from the cellular level and the MSK system



Fascia

- Fascia is the connective tissue that unites all aspects of the body. While by definition it functions as packing tissue and organ system cushioning, its cellular composition suggests that it is also critical for metabolic activity and immune surveillance.
- There are four distinct layers of fascia:
 - Pannicular
 - Axial (appendicular)
 - Meningeal
 - Visceral

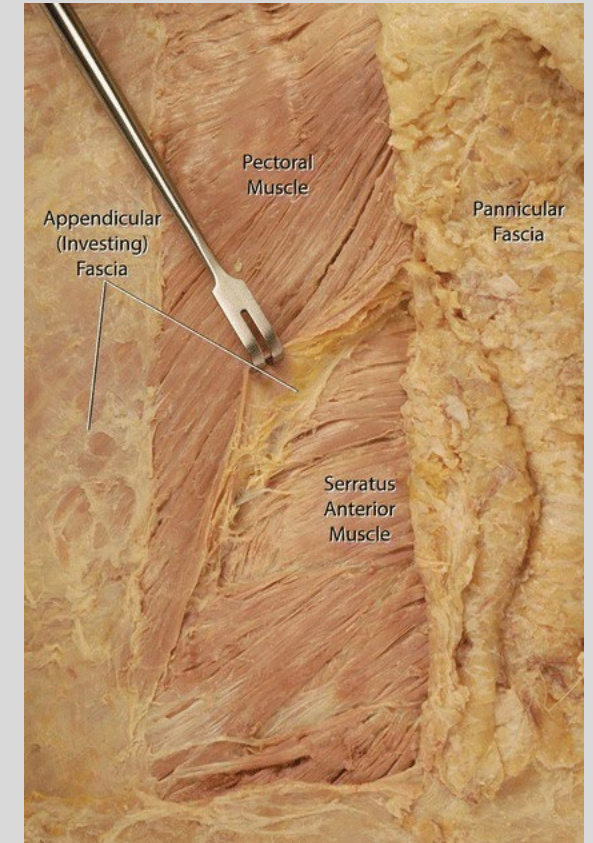


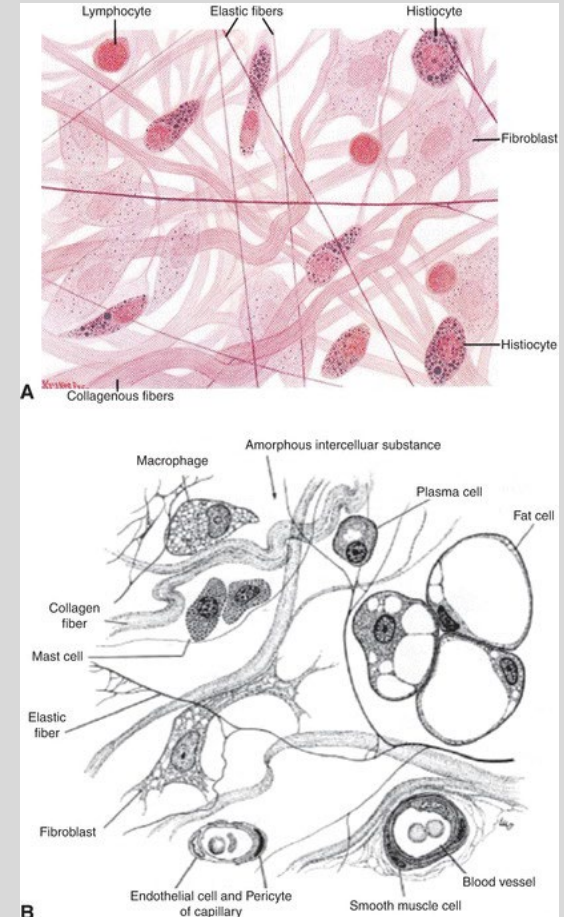
FIGURE 4.10 The fascia of the anterior chest wall. A window has been cut in the pannicular fascia as described in Figure 4.9. The inferolateral border of the pectoralis major muscle has been elevated to reveal the investing appendicular fascia separating this muscle from the underlying anterior serratus.

(Used with permission from the Willard/Carreiro Collection.)



Fascia

- Fibrocytes are the principal cell type of fascia; they synthesize and secrete: cytokines (e.g., interleukins [ILs], interferons, growth factors, etc.) and connective tissue extracellular matrix proteins (ECMPs), such as collagen, fibronectin, and others
- Strain frequency, magnitude, duration, and direction can influence the nuclear-mediated response of fibroblasts to the stimulus.



Mechanisms of OMM using fibroblast cell cultures

- JAOA • Vol 109 • No 5 • May 2009 • 258-260
- **Mechanical strain and the stimulation of cytokines are two important mechanisms leading to airway remodeling.**
- Bronchial and nasal fibroblasts differentially alter cytokine secretion in response to mechanical strain in patients with asthma vs healthy controls.
- No differences in cytokine secretion were observed among the fibroblast samples under unstrained control conditions.
- After 24 hours' strain, asthmatic bronchial fibroblasts secreted more IL-6 than normal controls.
- **Strain resulted in IL-6 upregulation and increased message for extracellular matrix protein in bronchial fibroblasts from asthmatic patients only**, and may reflect a propensity among these patients for airway remodeling.

Fascia thickness, aging and flexibility: is there an association?

Jan Wilke¹, Veronica Macchi², Raffaele De Caro², Carla Stecco²

Affiliations + expand

PMID: 30417344 PMCID: [PMC6284431](#) DOI: [10.1111/joa.12902](#)

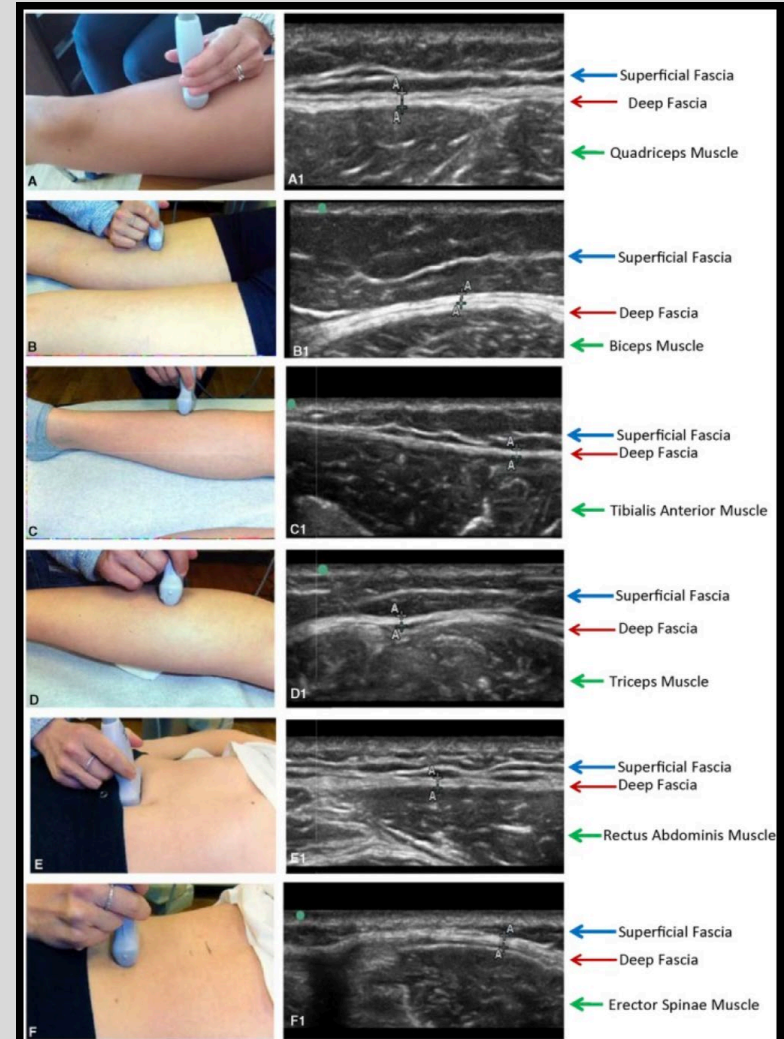
[Free PMC article](#)

Abstract

The morphology of the connective tissue may play an important role in locomotor mechanics. Recent research has revealed an association between increased fascia thickness and reduced joint flexibility in patients with chronic pain. The present study aimed to examine the relationship of both factors in healthy individuals, additionally testing the hypothesis that older subjects display a higher fascia thickness. Young ($n = 18$, 22 ± 1 years) and old ($n = 17$, 69 ± 4 years) healthy females were recruited for a quasi-experimental, cross-sectional trial. All participants underwent standardized ultrasound-based thickness measurements of the deep fasciae of the trunk and lower limb. Flexibility was assessed using sit and reach testing (hamstring extensibility) and the Schober test (lumbar flexion and extension). Systematic between-group differences of fascia thickness and variable associations (i.e. fascia thickness and flexibility) were detected using non-parametric data analyses. Young adults exhibited higher fascia thickness of the anterior and posterior lower leg, anterior thigh and abdominal wall ($+12.3$ – 25.8% , $P < 0.05$). Conversely, older participants showed higher thickness in the lumbar spine ($+40.0$ – 76.7% , $P < 0.05$). Correlations of both body mass and fascia thickness ($\tau = 0.45$ – 0.75 , $P < 0.05$), as well as flexibility and fascia thickness ($\tau = 0.38$ – 0.42 , $P < 0.05$) were found. Age-related changes in fascia thickness may be a contributing factor of restrictions in joint range of motion. Further study delineating the cause-effect triangle of body mass index, flexibility and fascia thickness is necessary.

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Fascia connections



Spiral Line



Arm Line



Superficial Front Line



Deep Front Line



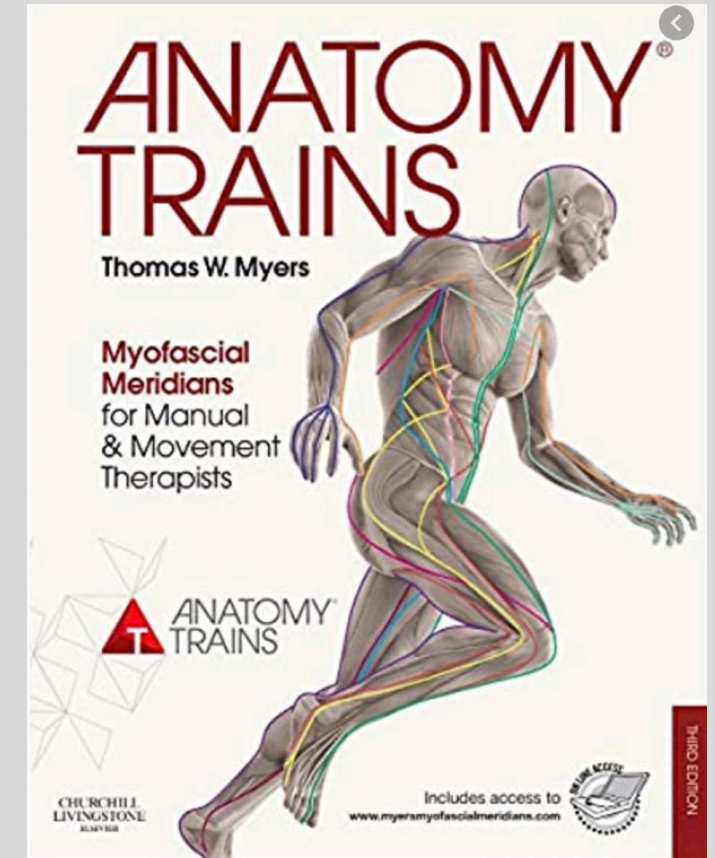
Lateral Line



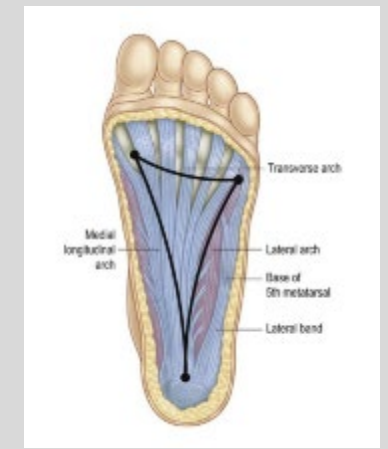
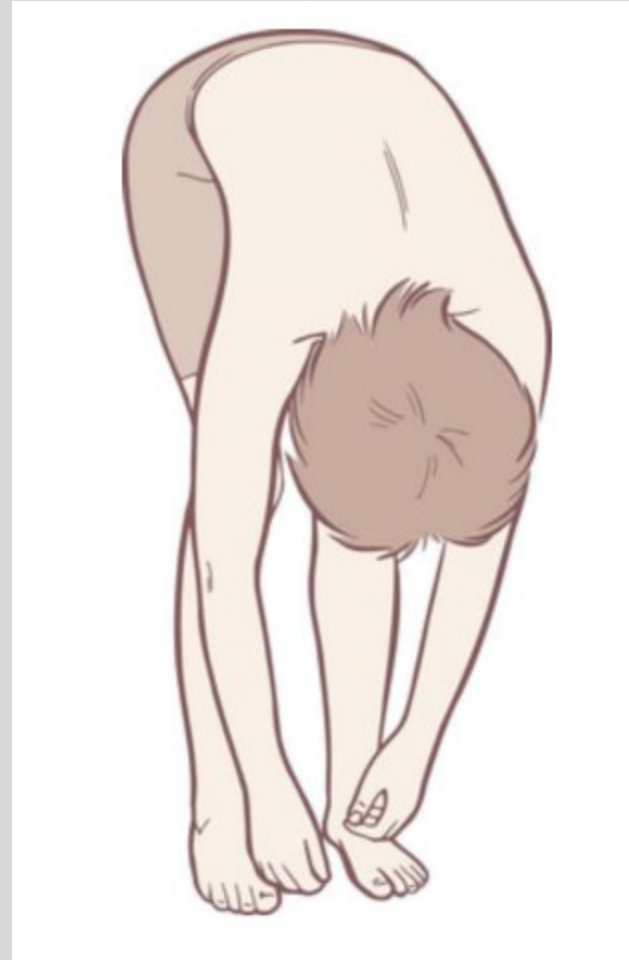
Superficial Back Line

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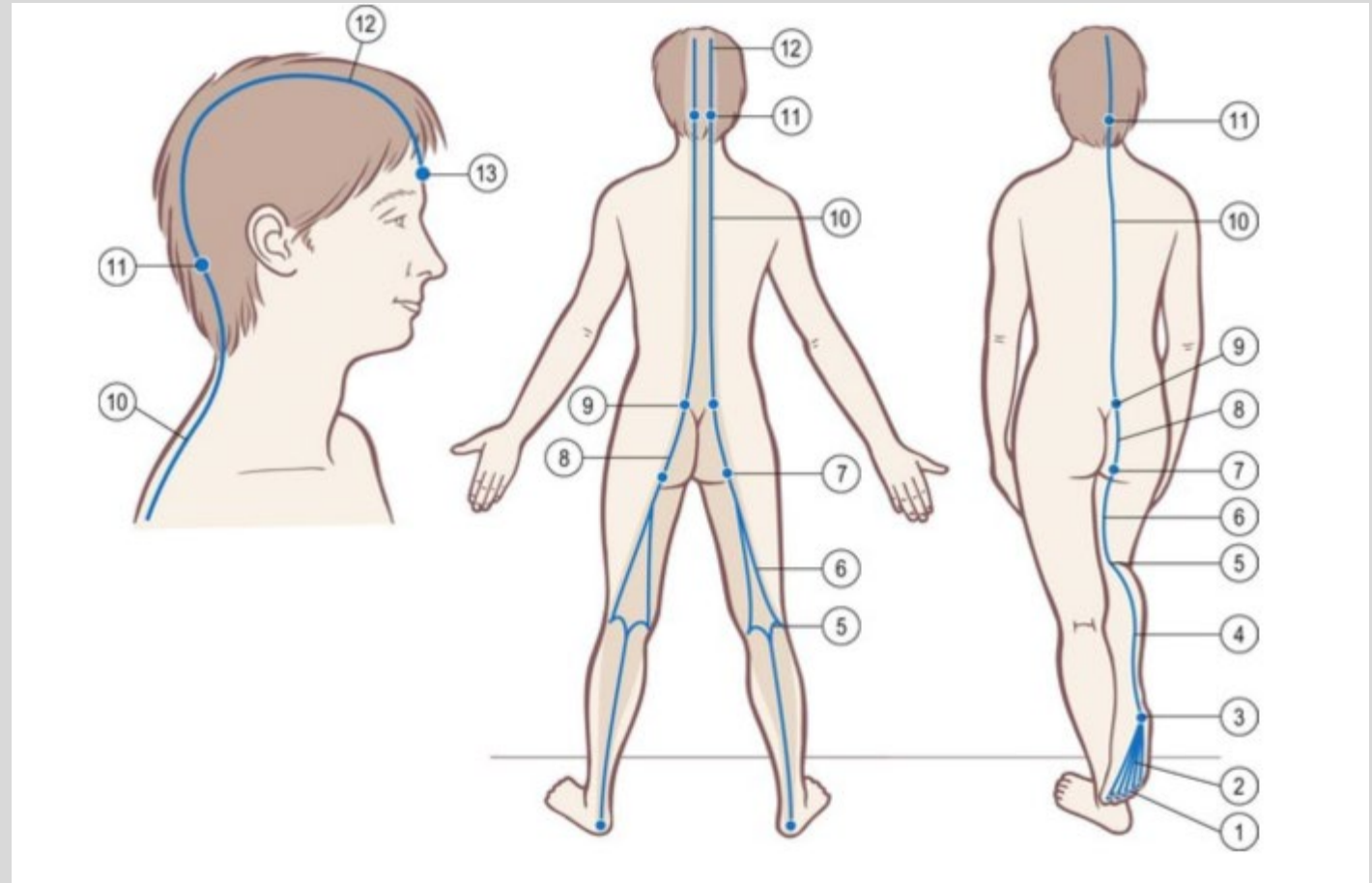
Superficial back line



Superficial back line

Superficial Back Line: myofascial 'tracks' and bony 'stations' (Fig. 3.2)

Bony stations		Myofascial tracks
Frontal bone, supraorbital ridge	13	
	12	Galea aponeurotica/epicranial fascia
Occipital ridge	11	
	10	Sacrolumbar fascia/erector spinae
Sacrum	9	
	8	Sacrotuberous ligament
Ischial tuberosity	7	
	6	Hamstrings
Condyles of femur	5	
	4	Gastrocnemius/Achilles tendon
Calcaneus	3	
	2	Plantar fascia and short toe flexors
Plantar surface of toe phalanges	1	



<https://basicmedicalkey.com/the-superficial-back-line/>

How does this information apply to Osteopathy?

- As osteopathic physicians we have the additional hands-on training to diagnose and treat physical dysfunctions that can influence these disease.
- Our treatment of the body has far reaching effects beyond just the musculoskeletal system but all the way down to the cellular level.
- You can treat many different disease processes by treating somatic dysfunction.

Where do we start?

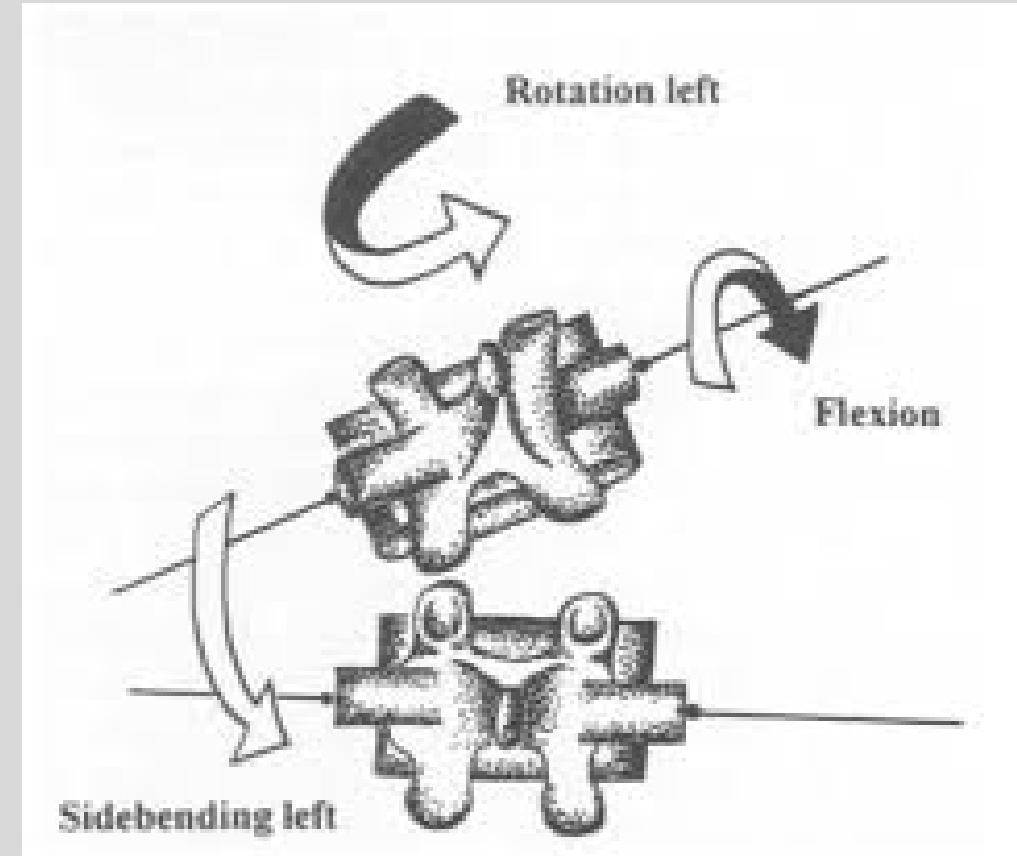


Area of Greatest Restriction & Sequencing

- Gives a starting place and an order to treatment.
- Allows the physician to focus in on the root of the problem and not just the areas of compensation.
- Reminds us to keep the whole person in mind during treatment and not to just focus on the area of complaint.

Fryette's Principle III

- If A Motion Is Introduced Into The Spine, The Other Two Motions Become Restricted
- If A Second Motion Is Introduced, The Other Two Motions Become Restricted.
- When A Third Motion Is Introduced, You Have A "Physiological Locking"
- Utilizing The Elastic Barrier to identify areas of restrictions



<https://www.sciencedirect.com/science/article/abs/pii/S1356689X98800258>

Application & Findings

Normal Joint

- You are at the **elastic/physiological barrier**.
- There will be a **yield, spring**, and “**soft end-feel**” with segmental motion testing.
- You can spring the joint simultaneously in three planes of motion

Restrictive Barrier

- When you challenge a dysfunctional area:
 - There will be **guarding** and **loss of tissue yield**.
 - **Joint spring is lost**, and you palpate a “**hard end-feel**.”
- **Find the most restricted/dysfunctional area.**
- **Grade dysfunctional regions: I, II, or III.**

Sequencing: Examination

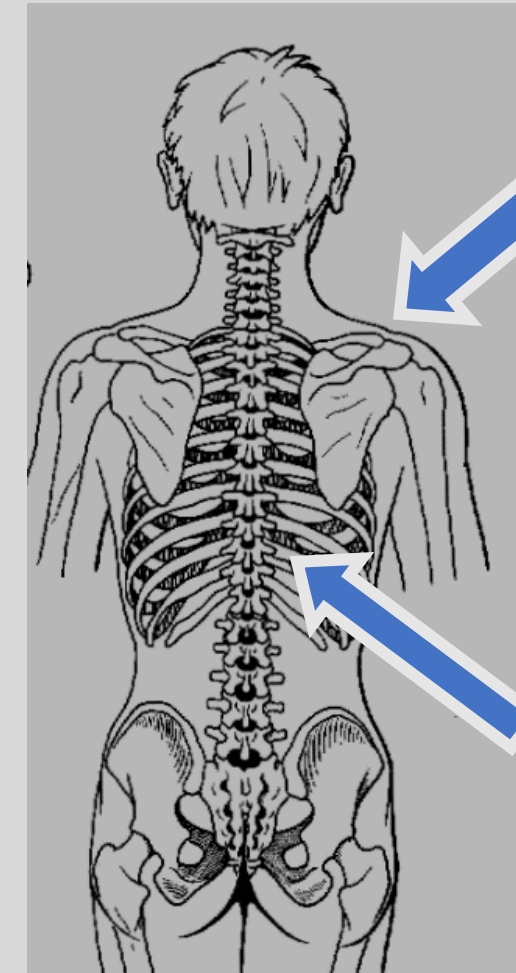
- Allow thumb to rest **lateral to spinous processes**, but **medial to transverse processes** — over the **facets** of all the cervical vertebrae.
- While in **neutral**, introduce **side bending** (*Type I*).
- Then slowly introduce **flexion/extension**, with each introduced movement until it segmentally moves down to the **base of the thumb** (*Type II*).
- Maintain this flexion. Now add **rotation** until movement segmentally moves down to the **base of the palpating thumb** (*Type II Enhanced*).
- **Findings:**
 - Normal: Soft & Springs
 - Dysfunctional: Hard & Lack Of Spring



Sequencing: examination

During The Palpatory Examination, Use Testing Movements As Follows:

- One Hand Introduces Movement.
- Other Hand Monitors The Segmental Response (“Listening Hand”)
 - Joint Mechanics
 - Parasegmental Tissue Response
- In Neutral, Introduce sidebending first, then flexion and extension, then rotation
- Slow “Sweeping” Movements Utilized
- Deliberate Motion: Interpreting The “End-Feel”
- Firm But Gentle Pressure
- Use the thumb or fingers on the listening hand to segmentally spring the dysfunctional area, in order to find the A.G.R. within the region. Identify the key within the dysfunctional area.
- Examine Both Standing & Sitting
(if key restriction is sacral - sitting → SI joint, if standing → gait dysfunction.)



OPERATOR'S
“MOTION
HAND”
(HAVE FINGER
TIPS OVER
CLAVICAL &
UPPER RIB
CAGE)

OPERATOR'S
PALPATING,
“LISTENING” &
“CHALLENGING
HAND”



USE SWEEPING MOTIONS **(TENSEGRITY MODEL)**

- **SLOW**
- **DELIBERATE**



INITIALLY INTRODUCE

- **SIDEBENDING IN NEUTRAL**
(INITIATING TYPE I MECHANICS)
- **THEN ADD F/E**
(ACTIVATES TYPE II MECHANICS)
- **THEN ADD ROTATION**

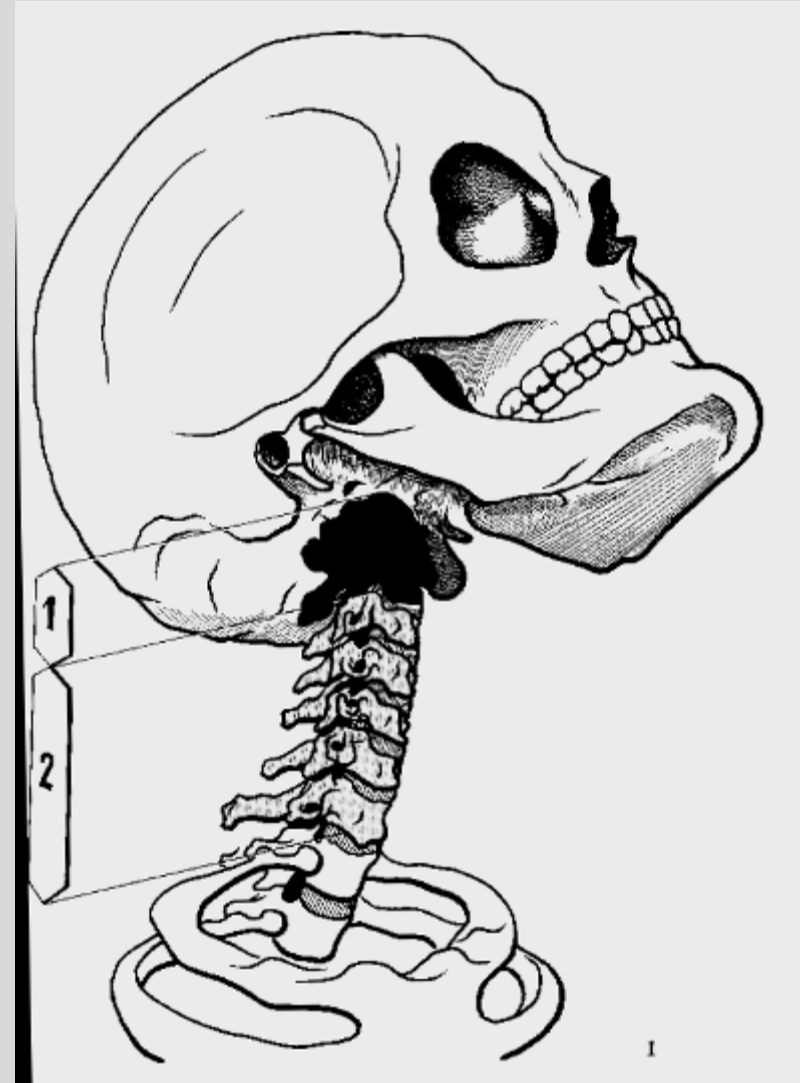
REPEAT OVER

- **LUMBAR AREA**
- **S/IJ's**



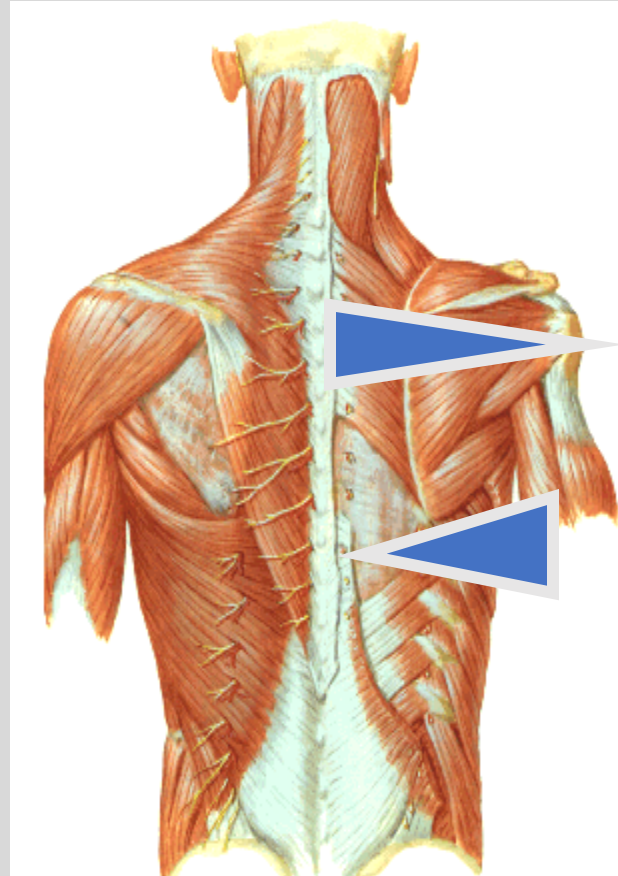
Determining AGR

- First comparing upper vs lower half of the body (cut off at thoracic diaphragm).
- If upper half of the body is the AGR Cranial, Upper or Lower Cervical



Thoracic Vs Rib AGR

- GET BETTER AS
PALPATE
LATERALLY?
- GET WORSE AS
PALPATE
LATERALLY?

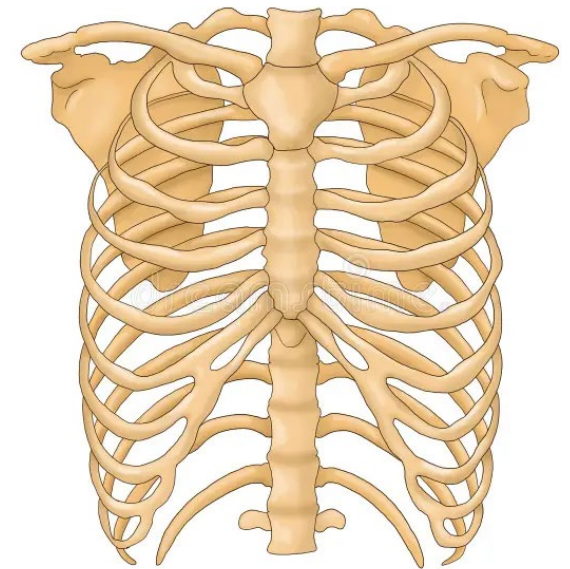


VERTEBRAL
PATTERN

RIBS CAGE
PATTERN

SEQUENCING: interpretation

- If the AGR is in the **rib cage**, it does **not** mean you must treat the specific rib used to make this decision.
- It only **points you toward evaluating the rib cage**.
- You must now **screen the rib cage** for the AGR
- Identify **Key group** and **key rib**.
- Respiratory Patterns: Find Key Rib
 - Treat the most restricted group if several present
 - Group dysfunctions treat key rib first
 - Key rib might be above or below
 - **B**otton **I**nhaled/**T**op **E**xhaled (B.I.T.E.)



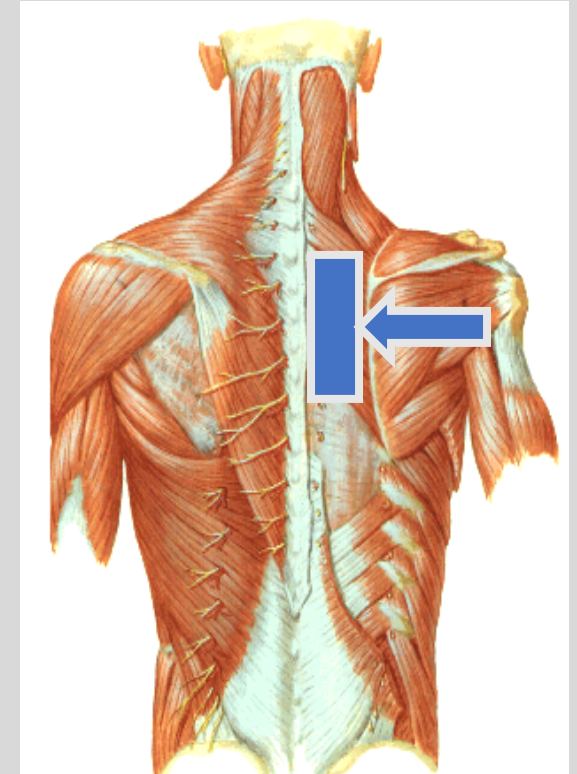
SEQUENCING: interpretation

Upper Extremity Pattern

- Diffuse medial scapular restriction
- **Unilateral pattern:** vertical band of restriction — *“Trap” and rhomboids tight on that side*

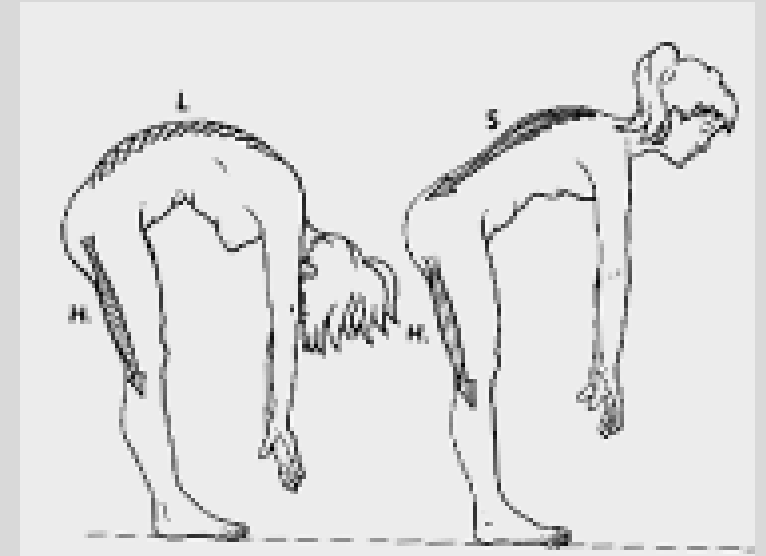
Screen for Dysfunction At:

- Sternoclavicular (S/C) and acromioclavicular (A/C) joints
- Radial head
- Wrist and hand
- You will frequently obtain **significant historical information** once this pattern is discovered!



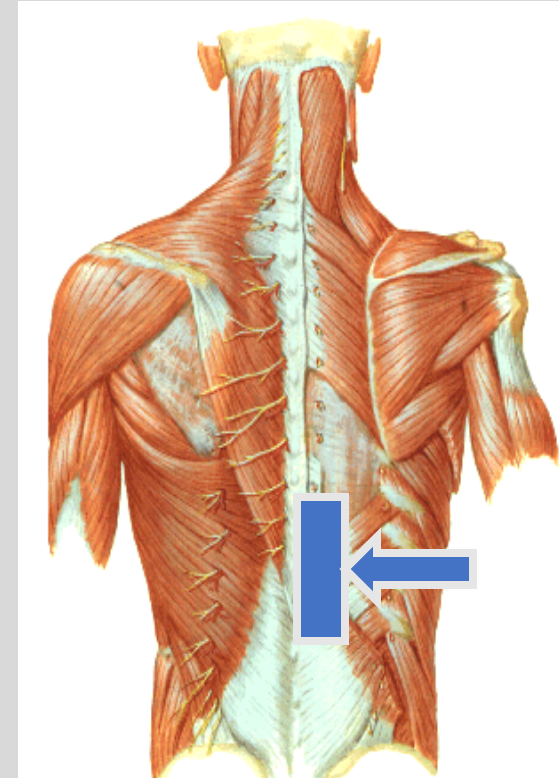
SEQUENCING: interpretation

- Lower Half of the Body Screening
- Standing Forward Bending Test = *Ilio-sacral*
→ Evaluates the **legs** and **innominates** on the sacrum.
- Sitting Forward Bending Test = *Sacro-iliac*
→ Evaluates the **sacrum** between the innominates.
- Lumbar Mechanics with FBT
 - Observe: Lumbar area: **lordosis**, **curvatures**
 - Hip rotation



SEQUENCING: interpretation

- If Unilateral Lumbar Pattern is present consider Lower Extremity Key restriction
- **Paravertebral tightness** is diffuse and limited to **one side** of the lumbar region.
- The **key** will usually be in the **ipsilateral leg**.
- **Standing FBT** will also be more dramatic than **sitting FBT**.
- **Screen for AGR in LE**
 - Fibular heads and tibial rotations
 - Ankle and foot
 - Hip restrictors
- When dysfunction is corrected, it usually **eliminates the unilateral lumbar pattern!**



UNILATERAL
PATTERN: LEG

SEQUENCING: interpretation

- **Key Concept**
- The **Area of Greatest Restriction (AGR)** — the **key** — may be, and frequently is, located in a **totally different area of the body** than the site of the **chief complaint**!
- The **pain** is frequently: Located where the body is **attempting to compensate**,
- **NOT** at the **AGR**.

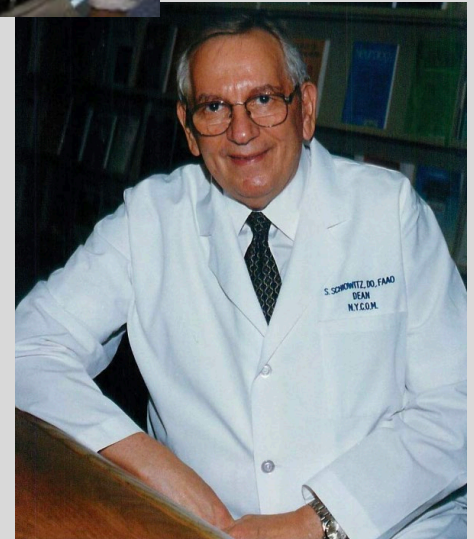
SEQUENCING: interpretation

- **Treatment Sequence**
- Once you find the **regional AGR**,
 - **Find the segmental AGR**
 - **Establish the segmental diagnosis**
 - **Start treatment there!**
- After treatment, **re-screen**.
- Then move to the **next regional AGR** and continue until the treated area releases **poorly** — **You're done for that visit.**

- "Find it, fix it, and leave it alone" - Andrew Taylor Still, the founder of osteopathic medicine
- This phrase encapsulates a core tenet of osteopathy, emphasizing the body's natural ability to heal itself once any structural impediments have been corrected.

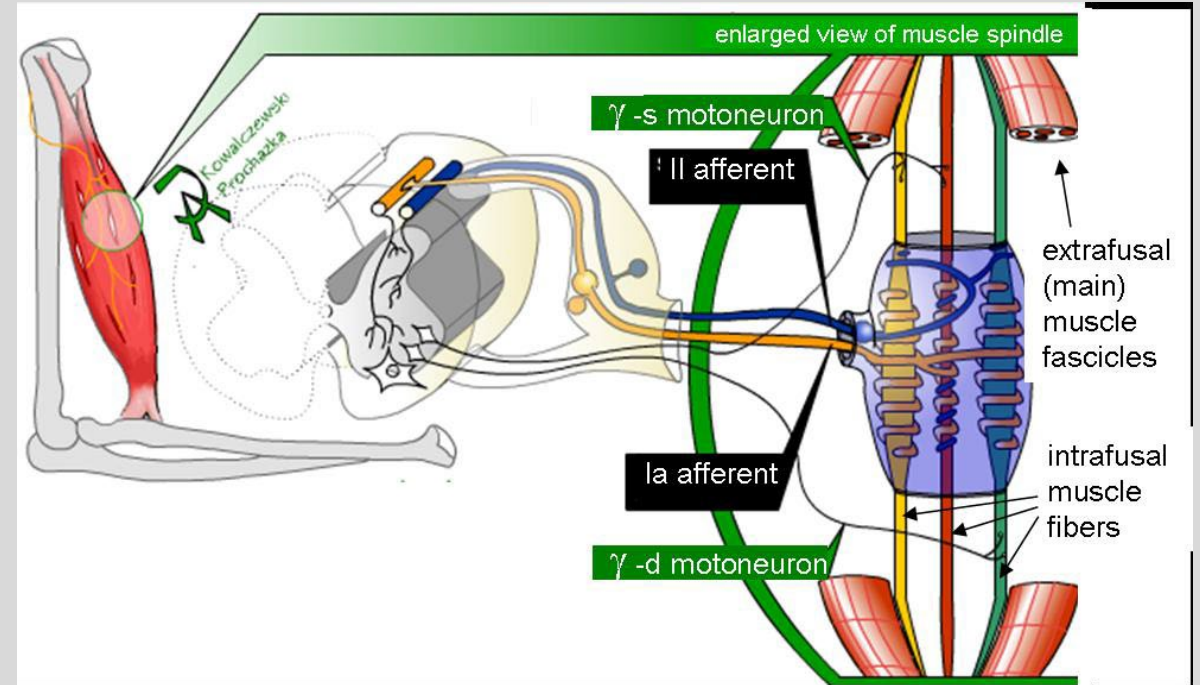
Facilitated Positional Release (FPR) Introduction

- Founder - Stanley Schiowitz, DO, FAAO
- Developed FPR in the 1990s
- Aimed to provide a rapid, effective method for relieving muscle tension and somatic dysfunction
- Influenced by earlier indirect techniques such as Strain-Counterstrain
- Designed for efficiency in clinical settings
- Commonly used in musculoskeletal medicine and pain management



Principles of FPR

- Indirect technique (moves the body away from restrictive barriers)
- Proposed Mechanism – sudden unloading of the muscle spindle results in a “quieting” of gamma motor gain to the muscle spindle, leading to a resetting of the tension and hypertonicity of the muscle fibers



Facilitated Positional Release (FPR)

Indications

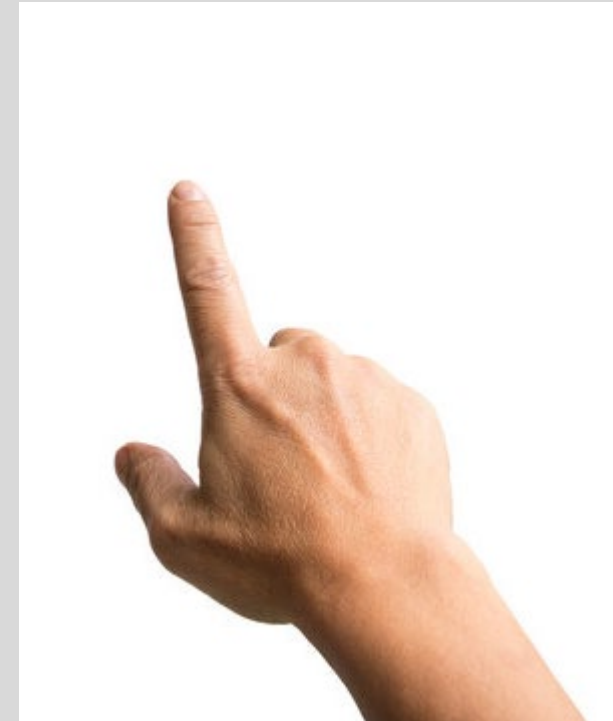
- Hypertonic muscle
- Somatic dysfunction (TART)

Contraindication

- Acute fractures or dislocations.
- Severe osteoporosis.
- Malignancy in the treatment area.
- Severe neurological deficits.
- Patient discomfort or inability to relax into the position.

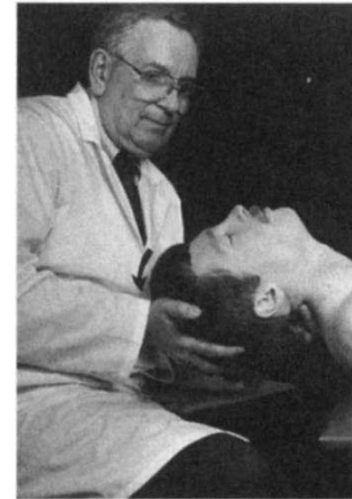
Key Steps for FPR

- Region of dysfunction placed in a neutral position
- Facilitating force applied (compression, torsion, or traction)
- Body part moved into position of ease
- Held for a few seconds (~3-5 seconds)
- Passive return to neutral
- Goal: Reduce muscle tension and improve range of motion
- Key is localization and being precise!
- Follow your monitoring finger!

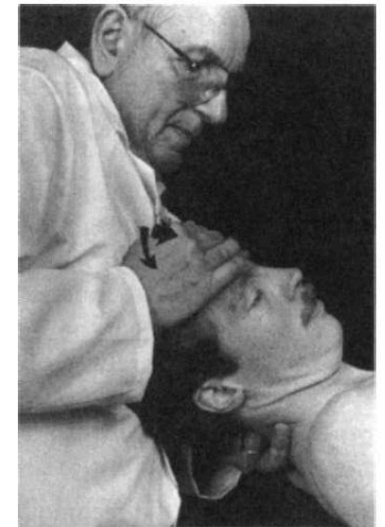


FPR - Cervical Spine – superficial muscle, Flexion/Extension SD

- **Positioning:** Patient supine, physician seated at head of table
- **Steps:**
 - Identify cervical segment with dysfunction
 - Neutralize region (reverse lordosis)
 - Apply axial compression
 - Move into position of ease (flexion/extension, side bending, rotation)
 - Hold 3-5 seconds, then release
 - Reassess



• FIG. 29-2 Facilitated positional release treatment of muscle hypertonicity in the cervical region with extension and right side-bending added.

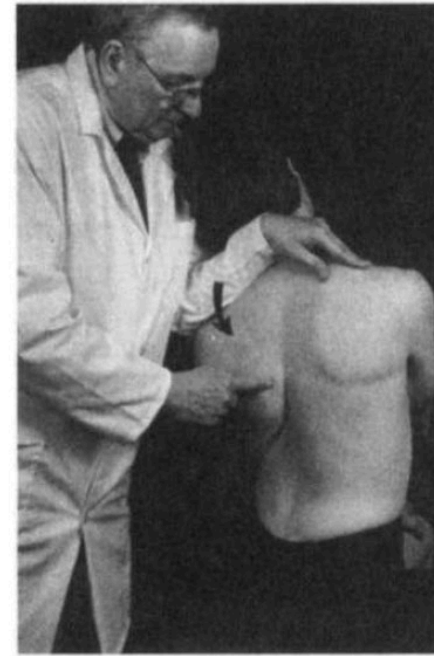


• FIG. 29-3 Facilitated positional release treatment for C4 flexion dysfunction with right side-bending and rotation.

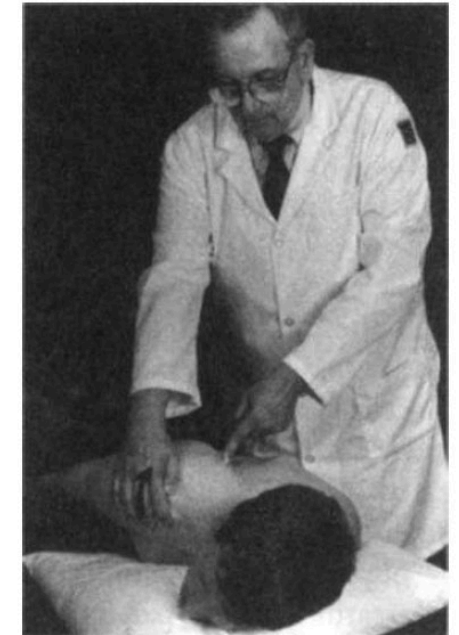
FPR - Thoracic Spine

- **Position:** Patient seated, physician standing on side of SD or on opposite side for prone position
- **Steps:**
 - Identify thoracic segment restriction
 - Adjust patient to neutralize spine
 - Apply gentle compression through shoulders to segment
 - Move into position of ease in all 3 planes
 - Hold 3-5 seconds, then release

DiGiovanna, E.L., Schiowitz, S. and Dowling, D.J. (2005) An Osteopathic Approach to Diagnosis and Treatment. 3rd Edition, Lippincott Williams & Wilkins, Philadelphia. Chapter 40



• **FIG. 40-2** Facilitated positional release treatment for thoracic superficial muscle hypertonicity; left side-bending added.

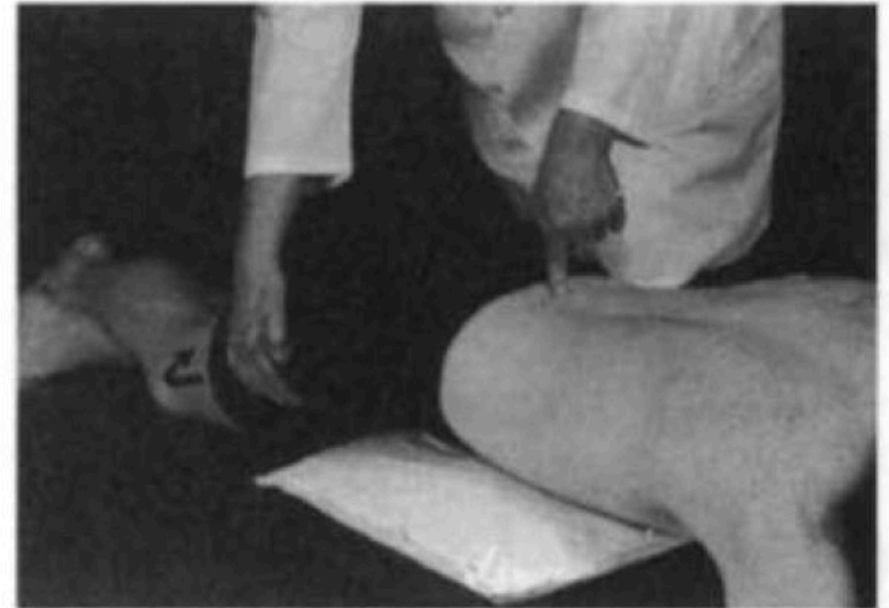


• **FIG. 40-4** A torsional motion is added, turning the shoulder upward and creating compression, side-bending, and rotation.

FPR - Lumbar Spine

Muscle hypertonicity

- **Patient Position:** prone
- **Steps:**
 - Identify lumbar hypertonicity
 - Pillow to neutralize lumbar lordosis
 - Standing on opposite side bring patients legs towards affected side to sidebend the pelvis over your finger.
 - Cross far leg over ipsilateral ankle.
 - Extend contralateral thigh and externally rotate to monitoring finger.
 - Hold 3-5 seconds, then release.



■ **FIG. 51-1** Facilitated positional release treatment for the left low back, superficial muscle hypertonicity.

DiGiovanna, E.L., Schiowitz, S. and Dowling, D.J. (2005) An Osteopathic Approach to Diagnosis and Treatment. 3rd Edition, Lippincott Williams & Wilkins, Philadelphia. Chapter 51

FPR - Lumbar Spine Extension SD

- **Patient Position:** prone
- **Steps:**
 - Monitor lumbar Extension SD
 - Pillow to neutralize lumbar lordosis
 - Standing on opposite side and extend and internally rotate the hip till motion is felt
 - Cross far leg over ipsilateral ankle.
 - Extend leg slowly.
 - Hold 3-5 seconds, then release.



• FIG. 51-2 Facilitated positional release treatment for a lumbar extension somatic dysfunction.

DiGiovanna, E.L., Schiowitz, S. and Dowling, D.J. (2005) An Osteopathic Approach to Diagnosis and Treatment. 3rd Edition, Lippincott Williams & Wilkins, Philadelphia. Chapter 51

FPR - Lumbar Spine Flexion SD

- **Patient Position:** prone
- **Steps:**
 - Monitor lumbar Flexion SD
 - Pillow to neutralize lumbar lordosis
 - Sit same side and flex and adduct and internally rotate the hip till motion is felt
 - Add compression superiorly to monitoring hand for 3-5 seconds, then release.



• FIG. 51-4 Facilitated positional release treatment for a lumbar flexion somatic dysfunction.

Summary

- Osteopathic screening through finding the area of greatest restriction is one approach to help target key areas to start treatment.
- Utilizing a systematic approach will help to keep the whole person and body regions in mind and prevent focusing on the patient complaint.
- Utilization of finding the area of greatest restriction can help you find the true key problem that is causing problems in your patients.
- Understanding fascial connections will help to improve your understanding of treating the body.
- FPR is a safe and effective osteopathic treatment approach to address somatic dysfunctions of the spine

Lecture Feedback Form:

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